

Advanced Telecommunication Technologies and Management

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MASTERS 2

**FACULTY OF ECONOMIC SCIENCES AND BUSINESS ADMINISTRATION
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Delay Analysis

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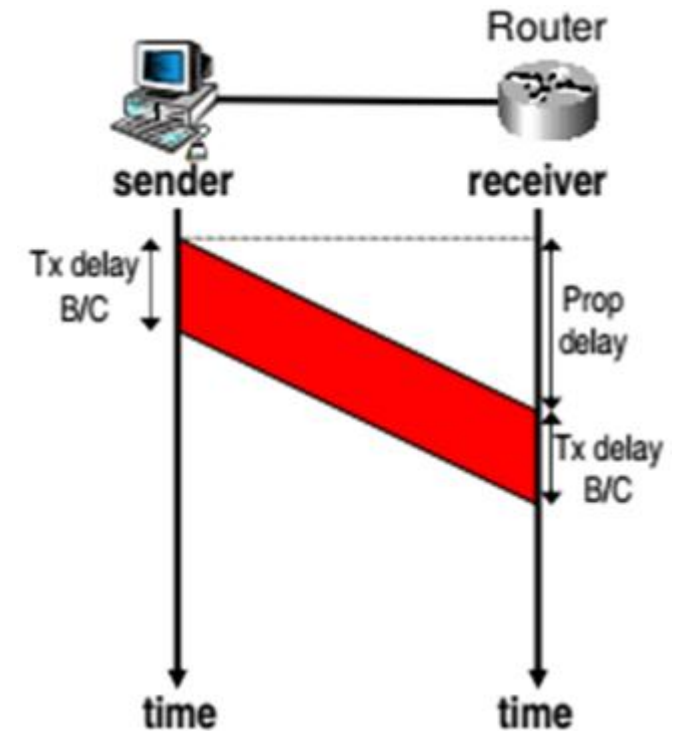
- Transmission Delay:

- C [bit/s] = Link rate
- B [bit] = Message size
- Transmission delay = B/C [sec]
 - ✦ 512 bytes packet
 - ✦ 64 kbps link
 - ✦ transmission delay = $512 \cdot 8 / 64000 = 64\text{ms}$

- Propagation Delay

- Link length
- Electromagnetic waves propagation speed in considered media
 - ✦ 200 km/s for copper links
 - ✦ 300 km/s in air

- Total delay = total propagation + total transmission+ total processing

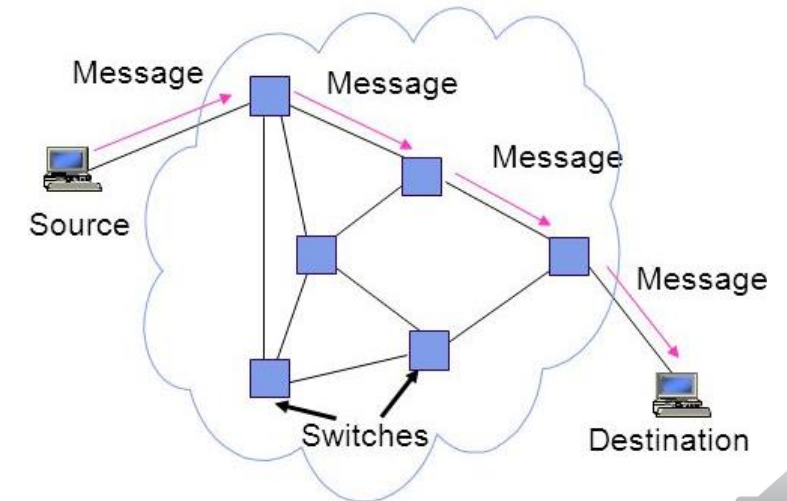


Switching Techniques

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- **Message Switching**

- No need to establish a dedicated path between two stations
- When a station sends a message, the destination address is appended to the message
- The message is then transmitted through the network, in its entirety, from node to node.
- Each node receives the entire message, stores it in its entirety on disk, and then transmits the message to the next node.
- Store & Forward Network
 - ✦ SMTP applications

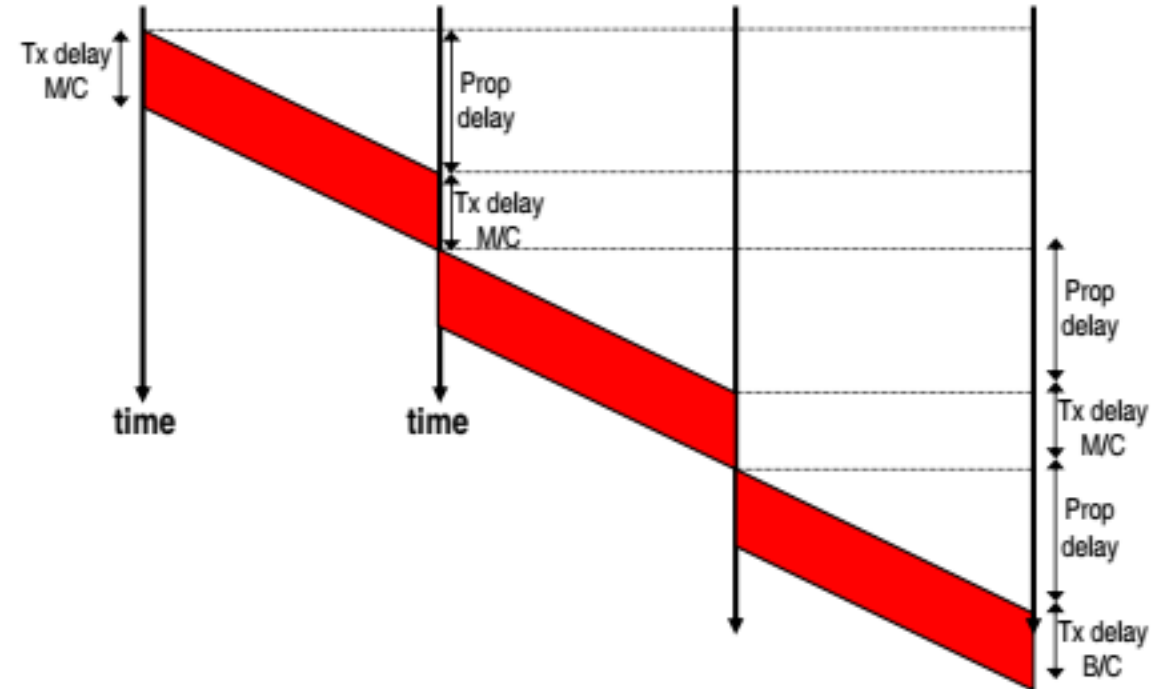


Delay Analysis – Message Switching

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- Assume:

- M = Number of hops
- P = Per-hop processing delay
- L = Link Propagation delay
- T = Packet Transmission delay



- Total delay = total propagation + total transmission + total processing = $ML + MT + (M-1)P$

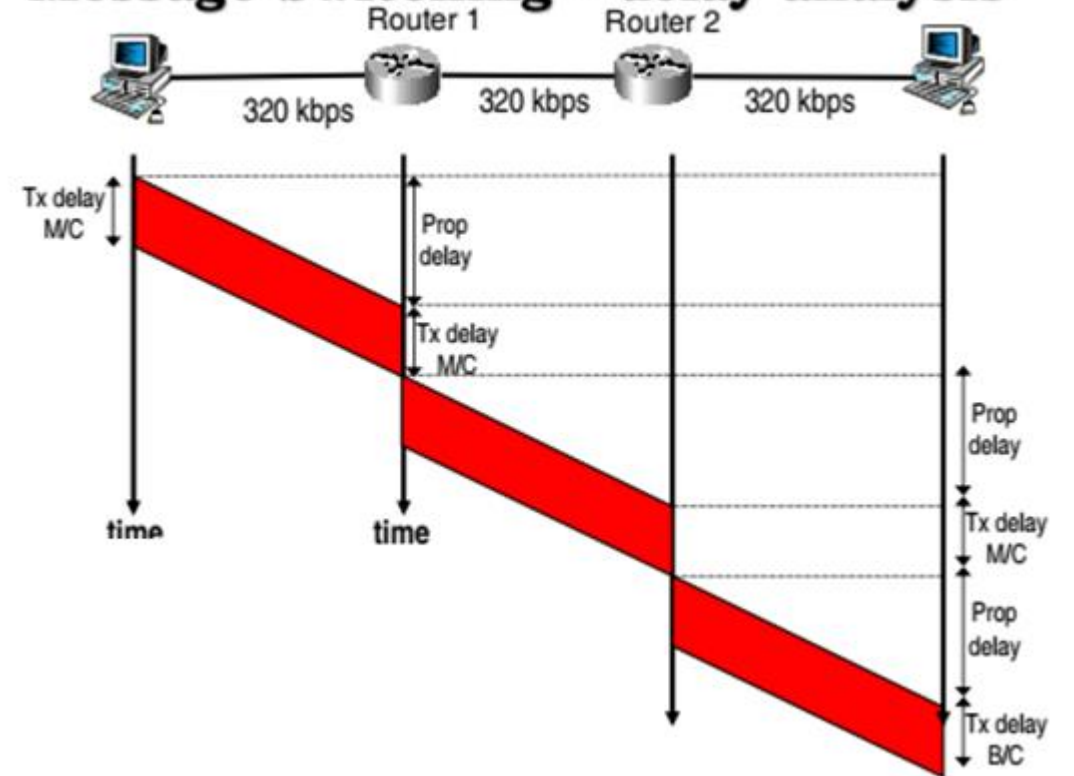


Delay Analysis – Message Switching

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- Exercise
- Message size = 400,000 bytes
- Header = 40 bytes
- Propagation $L = 0.050$ s

Message Switching – delay analysis

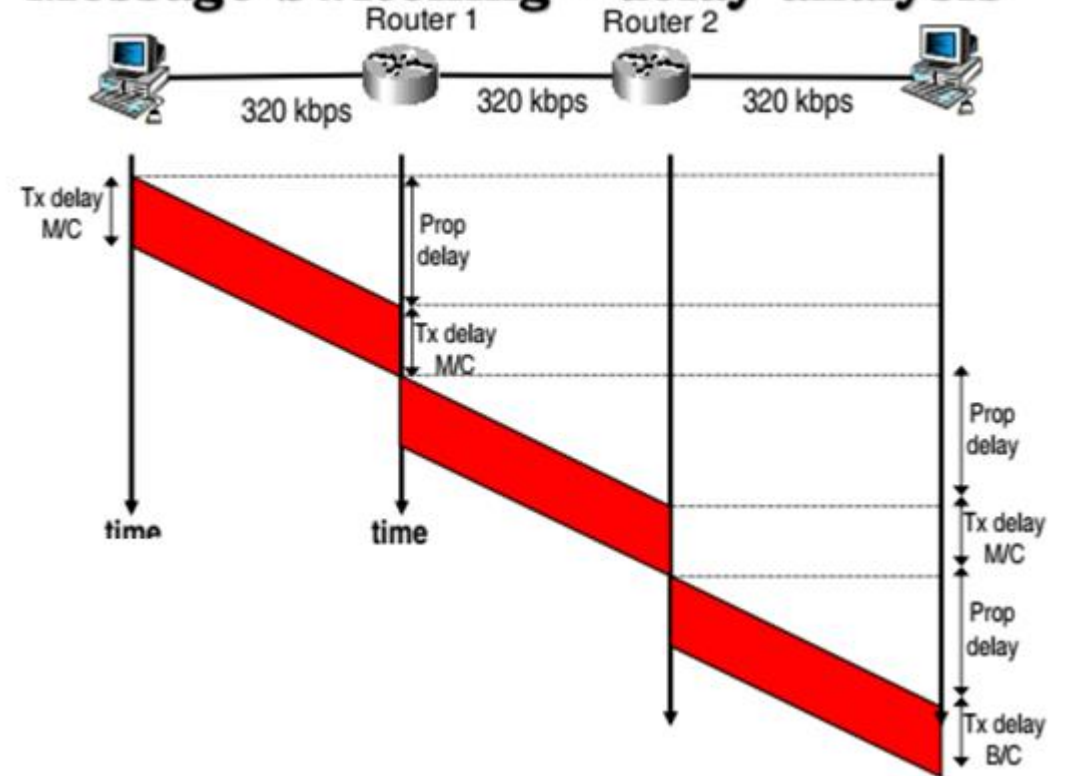


Delay Analysis – Message Switching

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- **Solution**
- Message size = 400,000 bytes
- Header = 40 bytes
- Propagation L = 0.050 s
- MsgHead = 400,040 bytes
- Delay = $ML + MT + (M-1)P$
 $= 3 * 0.05 + 3 * (\text{MsgHead} / C)$
 $= 30.153 \text{ s}$

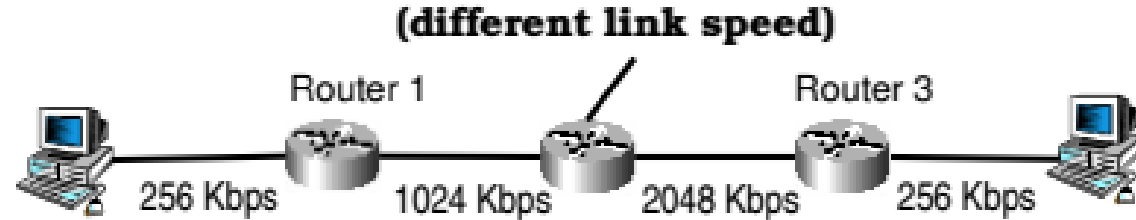
Message Switching – delay analysis



Delay Analysis – Message Switching

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- Exercise
- Time to transmit 1 MB file
- Assume 40 bytes header



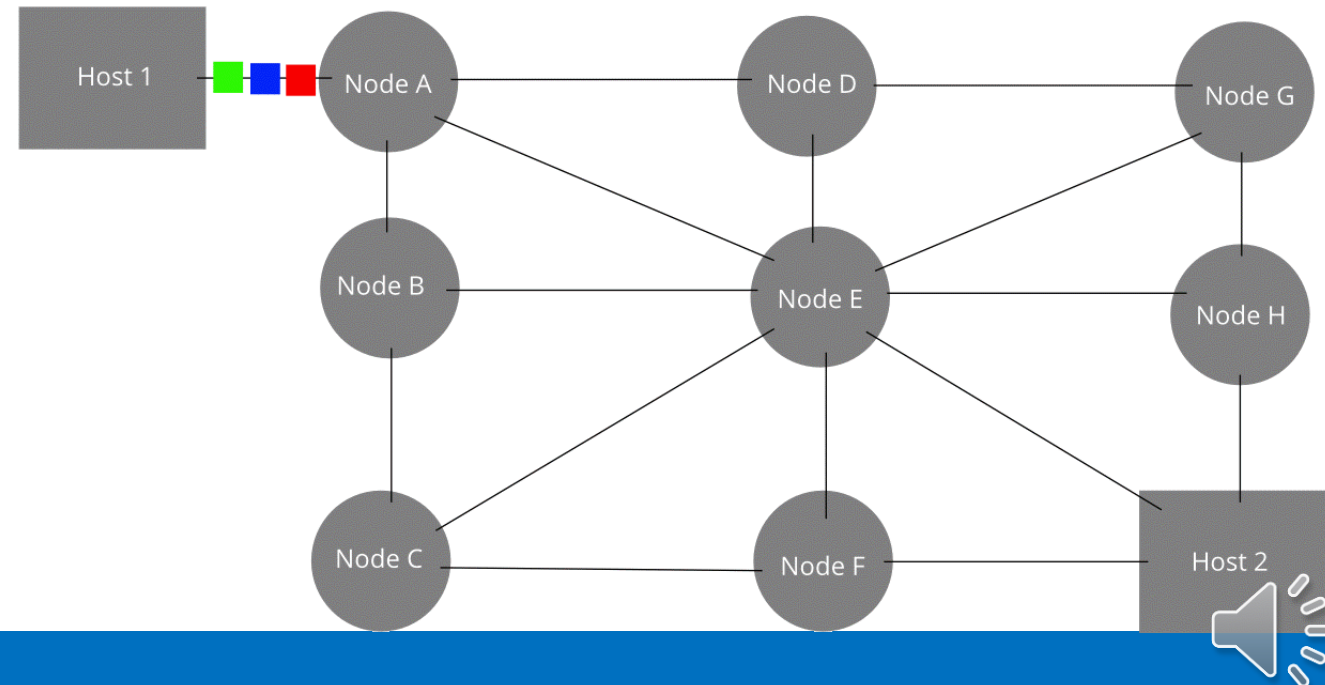
Switching Techniques

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- Packet Switching

- Data is sent in packets (header contains control info)
- Specifies a fixed size of packet which can be transmitted across network
- At the destination, all these packets has to be reassembled, belonging to the same file.
- Uses Store and Forward technique

The original message is Green, Blue, Red.



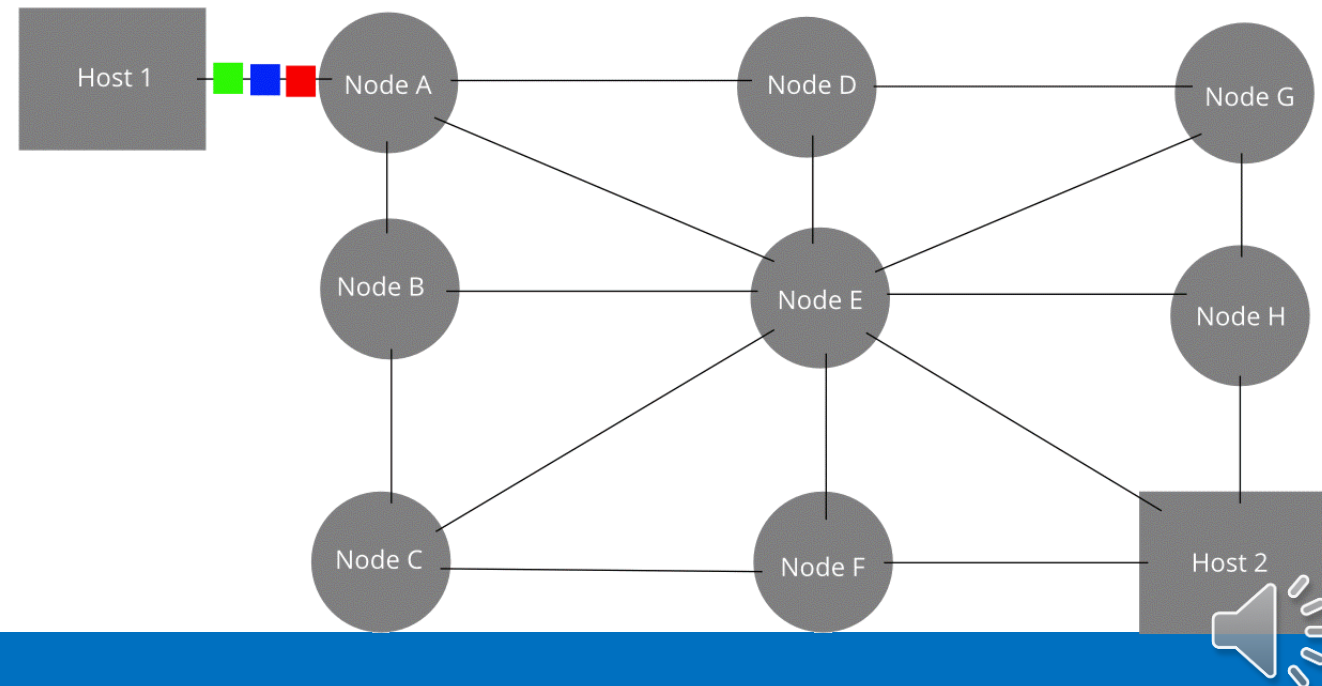
Switching Techniques

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- Packet Switching

- Each packet contains Source and destination address using which they independently travel through the network.
- Packets belonging to the same file may or may not travel through the same path.
- Throughput of network is improved
 - ✦ Internet Applications

The original message is Green, Blue, Red.



Switching Techniques

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- Advantage of Packet Switching over Circuit Switching :
 - More efficient in terms of bandwidth, since the concept of reserving circuit is not there.
 - Minimal transmission latency.
 - More reliable as destination can detect the missing packet.
 - More fault tolerant because packets may follow different path in case any link is down, unlike Circuit Switching.
 - Cost effective and comparatively cheaper to implement.



Switching Techniques

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- Disadvantage of Packet Switching over Circuit Switching :
 - Packet Switching don't give packets in order, whereas Circuit Switching provides ordered delivery of packets because all the packets follow the same path.
 - Since the packets are unordered, we need to provide sequence numbers to each packet.
 - Complexity is more at each node because of the facility to follow multiple path.
 - Transmission delay is more because of rerouting.
 - Packet Switching is beneficial only for small messages, but for bursty data (large messages) Circuit Switching is better.

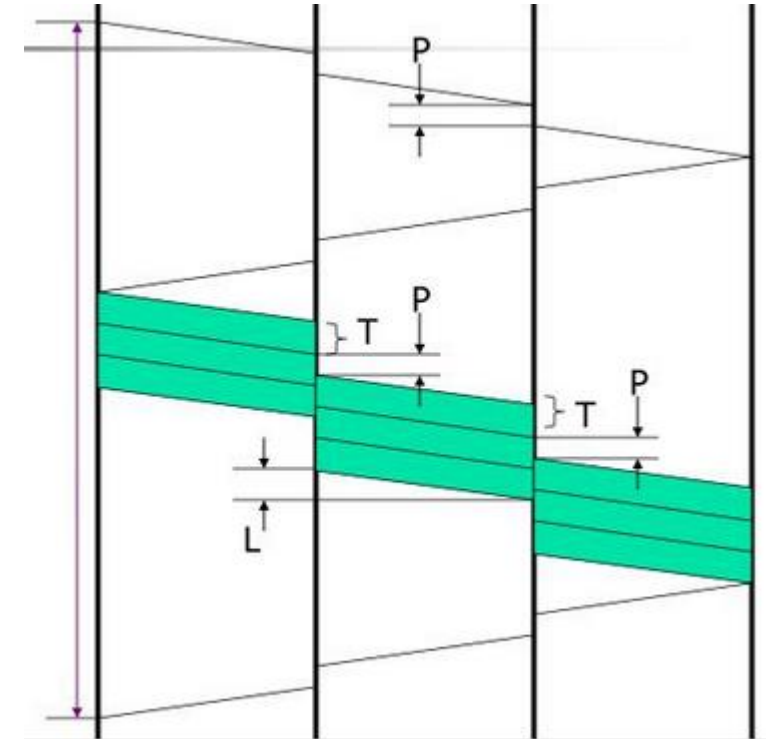


Delay Analysis – Packet Switching

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- Assume:

- M = Number of hops
- P = Per-hop processing delay
- L = Link Propagation delay
- T = Packet Transmission delay
- N = Number of Packets



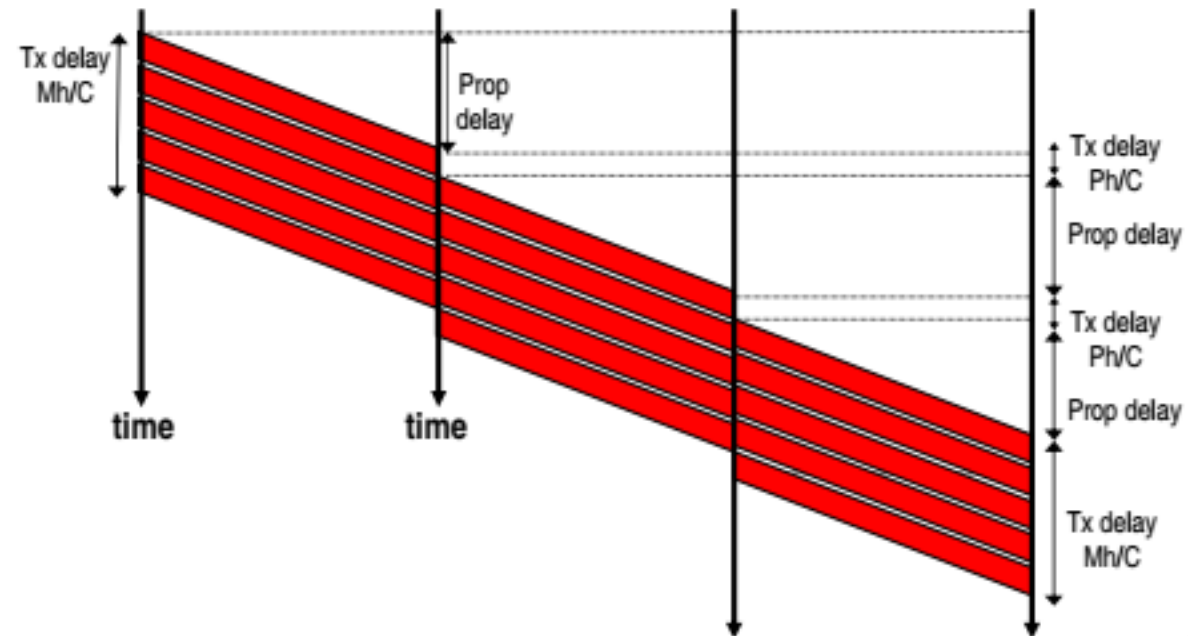
- Total delay = total propagation + total transmission+ total store&forward
+total processing = $ML + NT + (M-1)T + (M-1)P$



Delay Analysis – Packet Switching

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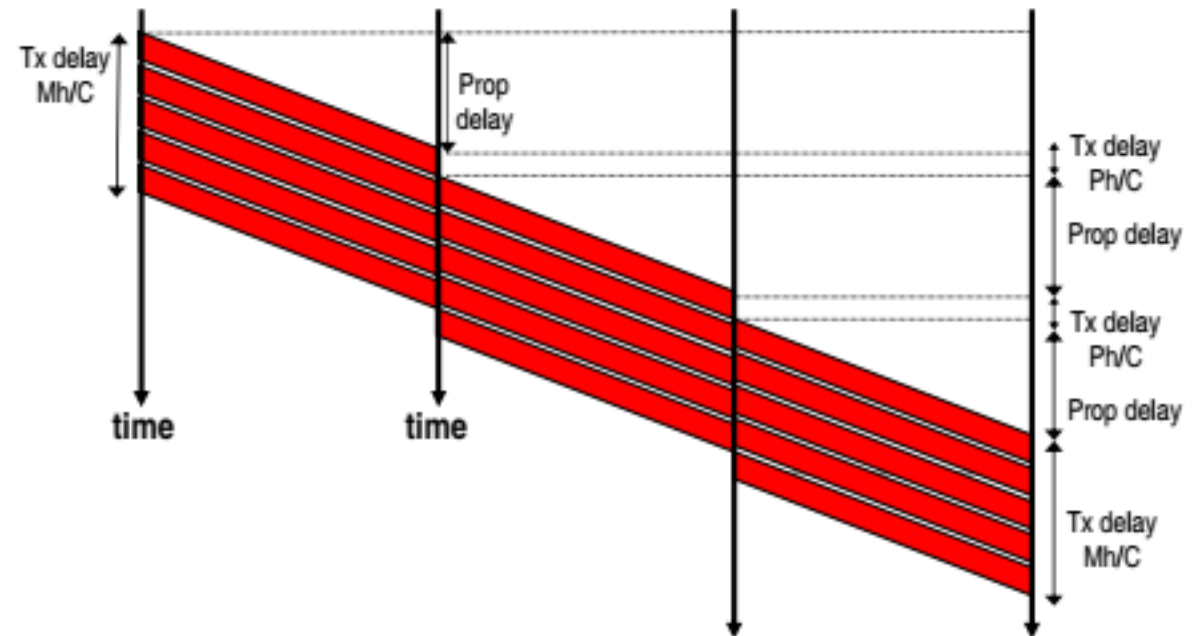
- Exercise
- Packet $P = 80,000$ bytes
- $H = 40$ bytes header
- Message size = $400,000$ bytes
- Propagation $L = 0.050$ s



Delay Analysis – Packet Switching

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- Solution
- Packet $P = 80,000$ bytes
- $H = 40$ bytes header
- Message size = 400,000 bytes
- Propagation $L = 0.050$ s
- $C = 320$ kbps
- $P_h = 80,040$, $T = P_h/C$
- $N = \text{Msg size}/P = 5$
- Delay = $ML + NT + (M-1)T + (M-1)P$
 $= 3L + 5P_h/C + 2P_h/C = 14.157$



Delay Analysis

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- Q1: It's 1989. Alice and Bob are 4 hops separated on a datagram packet-switched system where each link is 100 mile long.
- Hop handling deferral is $5\mu\text{s}$.
- Packets are 1500 bytes in length.
- All links have a transmission speed of 56kbit/s (original speed of Internet backbone interfaces in the 80s).
- The speed of light in the wire is approximately 125,000 miles/s.
- In the event that Bob sends a 10-packet message to Alice, how long will it take Alice to receive the message up to the last bit(measured from the time Bob starts sending)?



Delay Analysis

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- Answer: We know the following:

- Number of bounces $M=4$
- Number of packets $N=10$
- Per-hop handling delay $P=5\text{ }\mu\text{s}=0.000005\text{s}$,
- Link propagation delay $L = \text{distance/speed of light} = 100/125,000 = 0.0008\text{s}$
- Packet size= 1500 bytes = $1500*8=12,000$ bits,
- Packet transmission delay $T = \text{packet size/transmission speed} = 12,000/56000 = 0.214\text{s}$.
- Delay = $ML + NT + (M-1)T + (M-1)P = 0.0032 + 2.14 + 0.642 + 0.000015 = 2.785\text{s}$.



Delay Analysis

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- Q2: Alice and Bob 12 years later. All is the same, with the exception of that link transmission speed now is 1Gbit/s. How long will it take Alice to receive the message up to the last bit(measured from the time Bob begins sending)?



Delay Analysis

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- Answer: As before, we know the following:
 - Number of bounces $M=4$
 - Number of packets $N=10$
 - Per-hop handling delay $P=5\ \mu\text{s}=0.000005\text{s}$,
 - Link propagation delay $L = \text{distance/speed of light} = 100/125,000 = 0.0008\text{s}$
 - Packet size = 1500 bytes = $1500 \times 8 = 12,000$ bits,
 - Packet transmission delay $T = \text{packet size/transmission speed} = 12,000/10^9 = 12\ \mu\text{s}$.
 - Delay = $ML + NT + (M-1)T + (M-1)P = 3200 + 120 + 36 + 15 = 3371\ \text{ms} = 3.371\text{ms}$.
- Note that the total delay is dominated by the transmission delay which depends on link speed. A link with a higher transmission speed can reduce the delay significantly.



Course objectives – Part 1

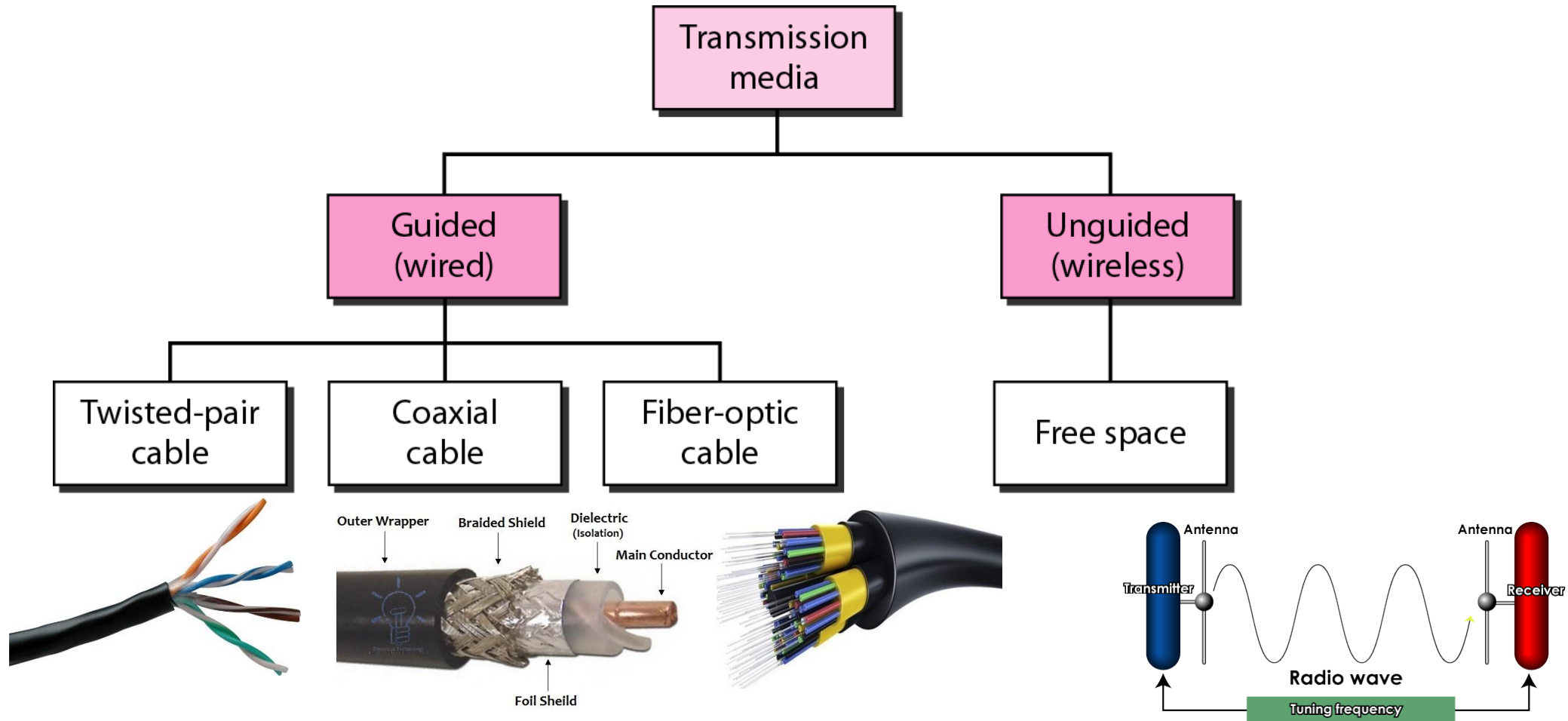
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- **Computer Communication Networks**
 - Introduction
 - Internet Protocols
 - Switching Techniques & Delay Analysis
 - **Transmission Medium & Allocation of Resources**
 - xDSL Technologies



Transmission Medium

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Transmission Medium

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- Comparison Table of Guided Transmission Medium

	Twisted Cable Pair	Coaxial Cable	Optical Fiber Cable
Data Transfer Rate	10Mbps-10Gbps	100 Mbps	More than 100 Gbps
Data Transfer Range	100m	185m-500m	-
Interference Susceptibility	More	Less than Ethernet cable	NIL
Cost of Cable	Least Cost	More than Ethernet	Very Expensive



Wireless and mobile communication networks

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- The devices used for wireless communication require certain features
 - Low power consumption
 - Light weight
 - Worldwide communication ability.
 - Access to a communication network is wireless so that the end users remain free to move.
- Limitations
 - Range of wireless communication is always limited.
 - Range of user mobility is also limited



Wireless and mobile communication networks

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- To overcome these limitations:
 - A geographical region is divided into smaller regions called CELLS.
 - Each cell has a fixed communication device that serves all mobile devices within that cell.
 - As a mobile device moves out of one cell and into another cell, service of that connection is transferred from one cell to another.
 - ✦ Handoff process.
 - Low power consumptions



Resource allocation techniques

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- Two approaches for allocation of resources among the users.
- Static allocation of resources
 - Quantity of resources is allocated to each user.
 - If user doesn't use his allocated resources, no one else can
- Dynamic allocation of resources
 - Desired quantity of resources is allocated to users on the basis of their demands and for the duration of their need.
 - Once the need is satisfied, the allocation is retrieved.
 - Someone else can use these resources if needed.



Challenges and Issues

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- Propagation Delay
- Integrations of different services in the same network & each type of services has its own requirements to achieve a desired level of quality of service
- Maintaining security and integrity of information.



Course objectives – Part 1

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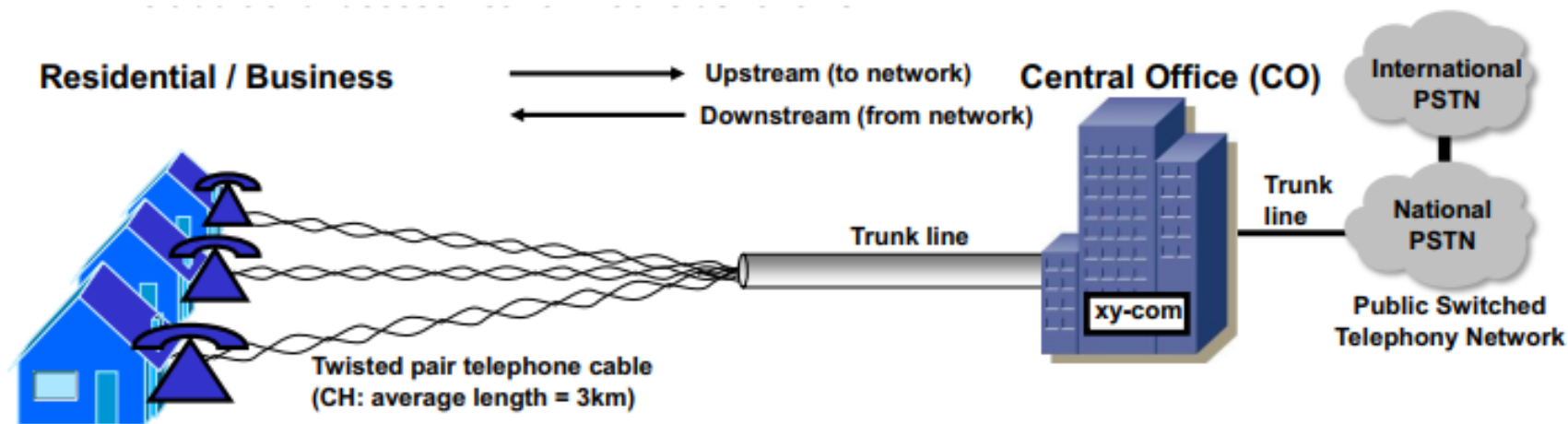
- Computer Communication Networks
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 - xDSL Technologies



xDSL Technologies

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- Digital Subscriber Line is a broadband high speed internet technology that provides digital data transmission over the existing copper wires of local telephone network.
 - services typically ranges from 256 Kb/s to 24 Mbit/s in the direction to the customer (downstream)
- The traditional access network looks as follows:

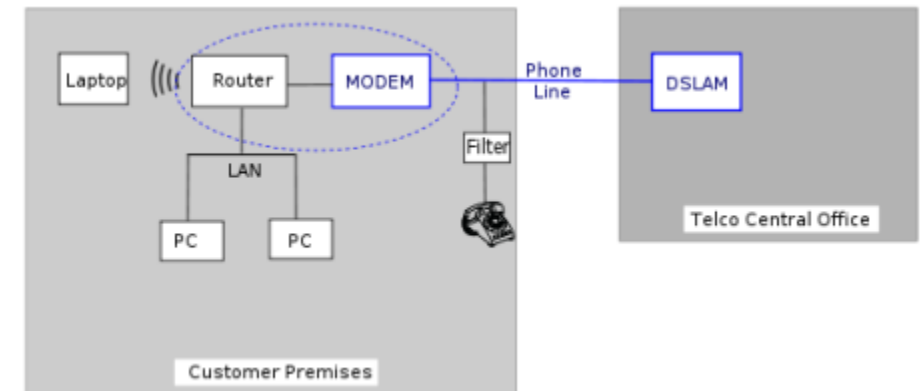
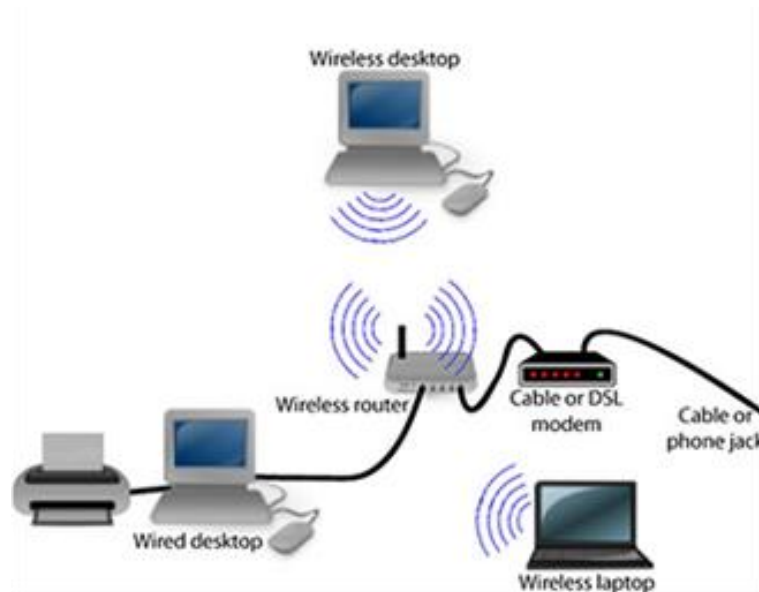


xDSL Technologies

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- DSL Requirements

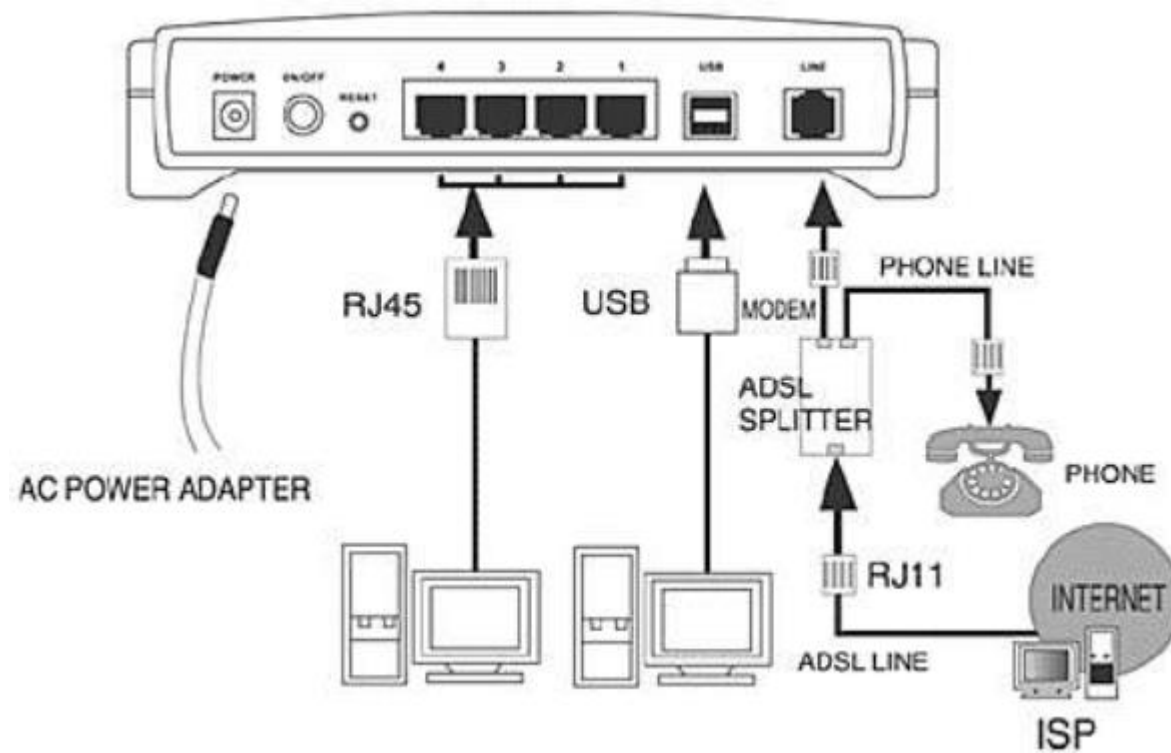
- Phone Jack
- Filter
- Modems
- Ethernet cable
- Telephone



xDSL Technologies

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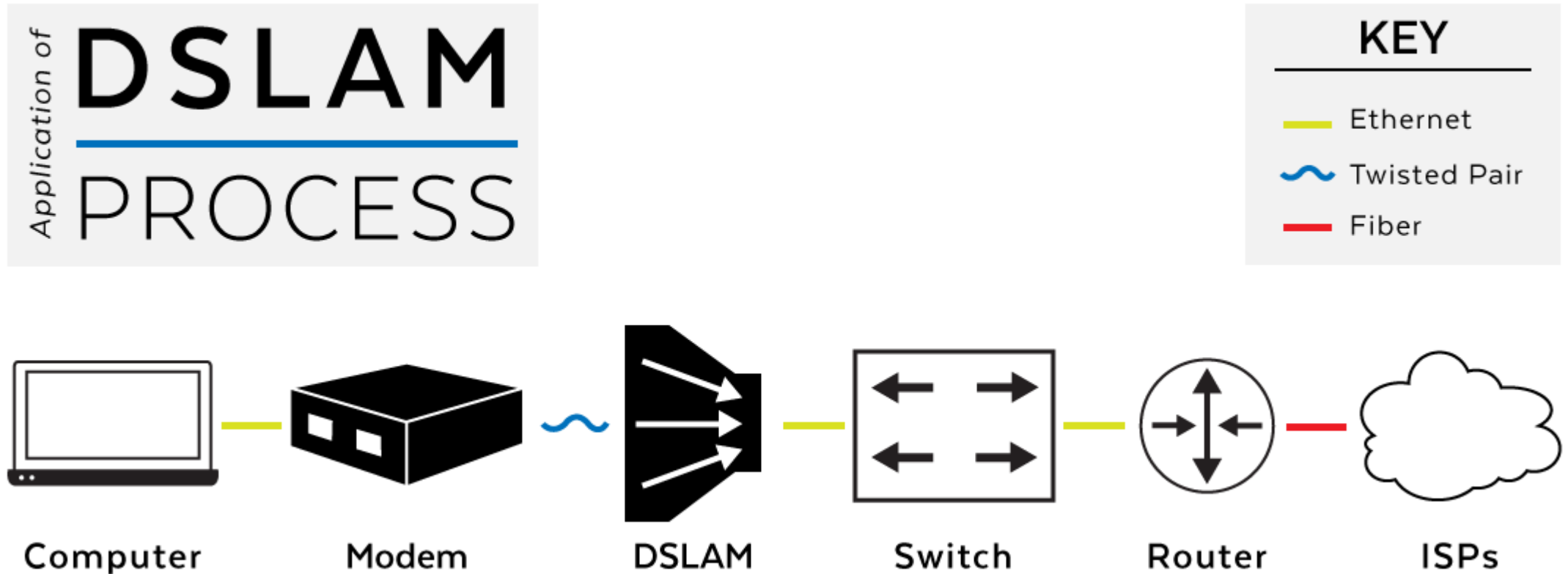
- DSL schematic



xDSL Technologies

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- DSL Requirements



- DSLAM, or Digital-Subscriber-Line-Access-Multiplexer
 - Network distribution device that aggregates individual subscriber lines into a high-capacity uplink.
 - These high capacity uplinks, either ATM or Gigabit Ethernet, connect subscribers to their Internet service providers (ISPs).
 - Typically located in telephone exchanges or distribution points.
 - Allow for the high-speed transmission of DSL technology using legacy copper lines.
 - Using advanced multiplexing techniques, these units salvage the utility of the millions of copper lines that were originally deployed for telephone usage.
 - DSLAMs also come with many advanced traffic management features to separate and prioritize voice, video, and data traffic.



- High Data Rate Digital Subscriber Line (HDSL / HDSL2), was the first DSL technology that uses a higher frequency spectrum of copper, twisted pair cables.
- Symmetric Digital Subscriber Line (SDSL / SHDSL), the volume of data flow is equal in both directions.
- Asymmetric Digital Subscriber Line (ADSL), the volume of data flow is greater in one direction than the other.
- Very High Speed Digital Subscriber Line (VDSL)
- Gigabit Digital Subscriber Line (GDSL)



xDSL Technologies

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- Main Characteristics

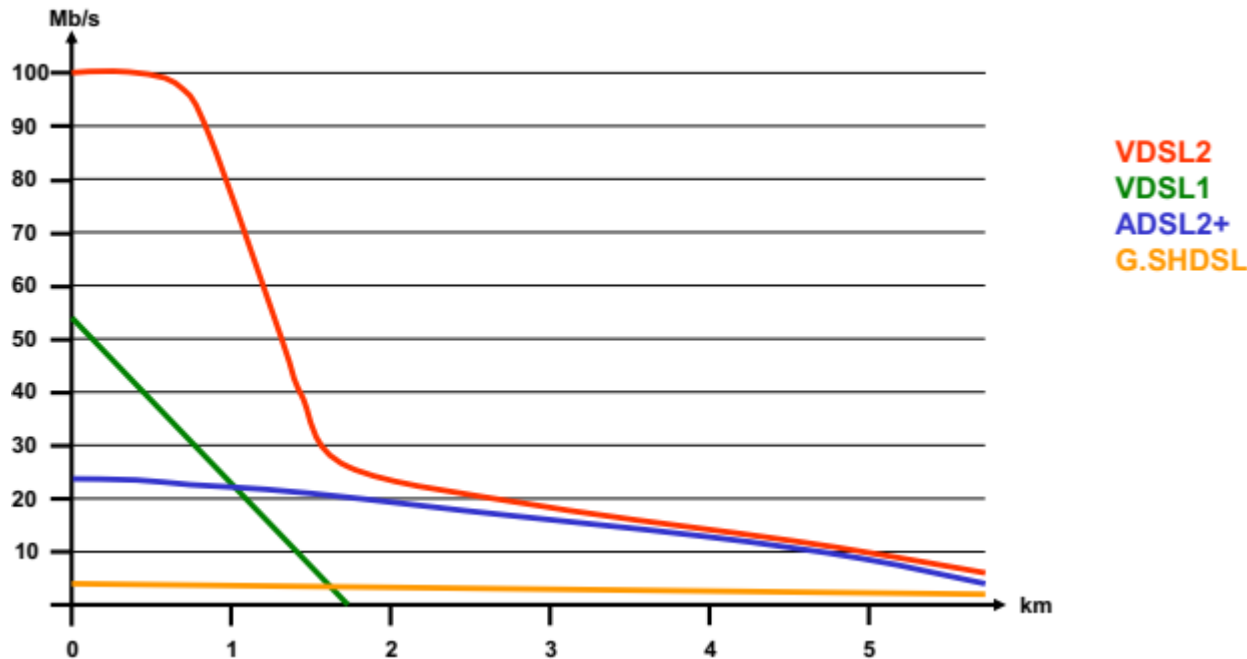
Name	Standard	Speed	Wires	Mode	Distance	Comment
ADSL (G.dmt)	G.992.1	1.5 - 9Mbps DS 16 - 640Kbps US	Single pair	Downstream & upstream	6km	Needs splitter
ADSL2+	G.992.5	...24Mbps DS ...3.3Mbps US	Single pair	Downstream & upstream	6km	Needs splitter
G.Lite=DSL Lite	G.992.2	1Mbps DS 128Kbps US	Single pair	Downstream & upstream	6km	Does not need splitter
HDSL		1.54Mbps	Two pairs	Duplex (symmetric)	5km	-
HDSL-II	G.991.1	1.54Mbps	Single pair	Duplex (symmetric)	5km	-
IDSL	I.430	144Kbps	Single pair	Duplex (symmetric)	6km	-
RADSL		1 - 7Mbps 128Kbps - 1.5Mbps	Single pair	Downstream & upstream	6km	Automatically adapts bit rate to line
G.SHDSL	G.991.2	192Kbps - 2.3Mbps	Single pair	Duplex (symmetric)	3.3km	No POTS/ISDN
VDSL1	G.993.1	13 - 52Mbps 1.5 - 2.3Mbps	Single pair	Downstream & upstream	1.2km	No POTS/ISDN
VDSL2	G.993.2	...100Mbps ...50Mbps	Single pair	Downstream & upstream	1.5km	No POTS/ISDN



xDSL Technologies

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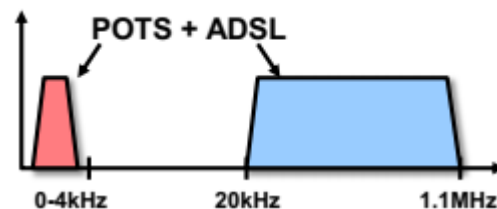
- Distance must be traded off against bandwidth (the higher the bandwidth the lower the range)



ADSL Technology

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- ADSL physical layer
 - Exploits the fact that (today's) Internet traffic is largely asymmetric (much higher traffic downstream = to user, low traffic upstream = from user).
 - Made to coexist with POTS
- Frequency usage of ADSL:



POTS' + ADSL's
frequencies are not
overlapping thus there
is no problem

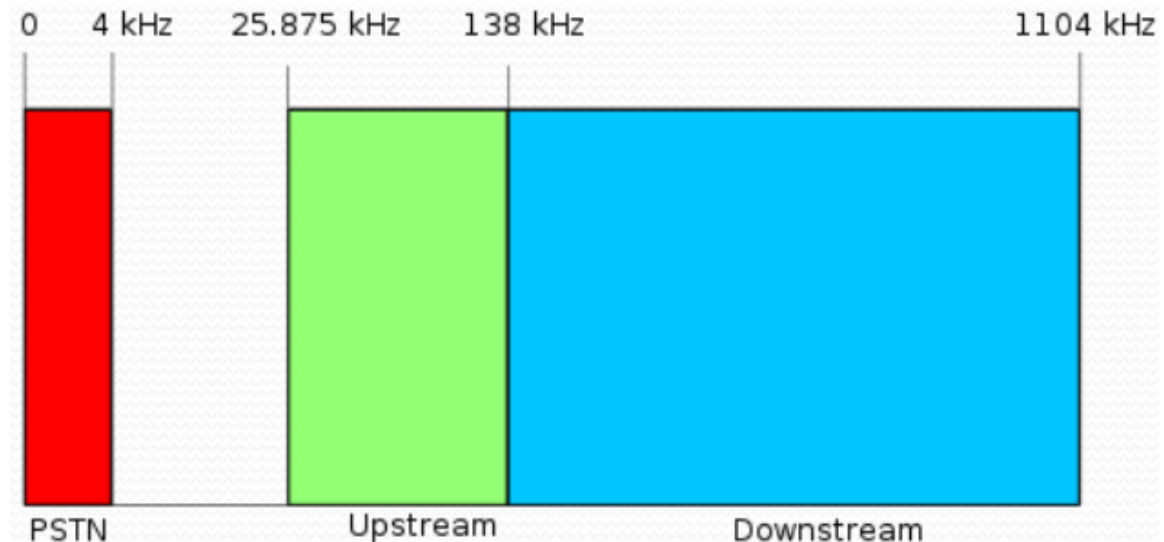


ADSL Technology

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- ADSL physical layer

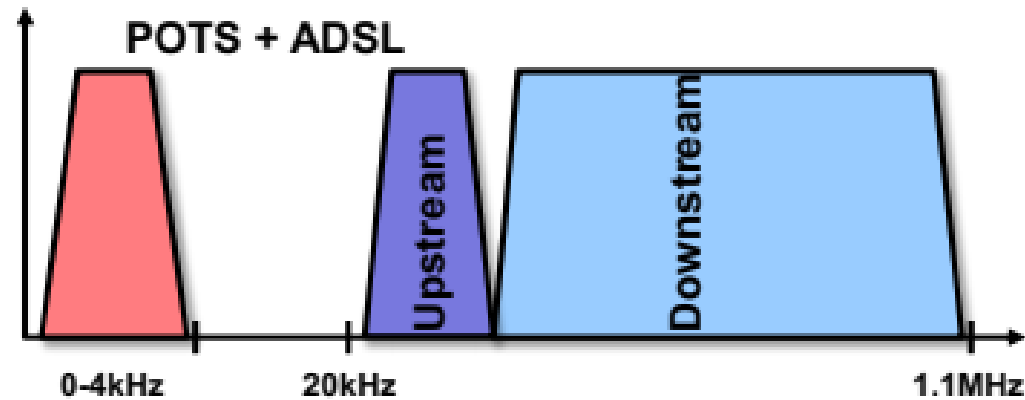
- Phase modulation coupled with amplitude modulation (QAM)
- Frequency spectrum from 0 thru 1.1MHz is divided into 256 sub-bands (channels 0-5 are not used - these frequencies are used for PSTN (POTS)).
- Sub-channels 6 thru 21 are used for upstream while sub-channels 22 thru 255 are used for downstream traffic.



ADSL Technology

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- ADSL physical layer
 - Upstream and downstream use non-overlapping frequency bands

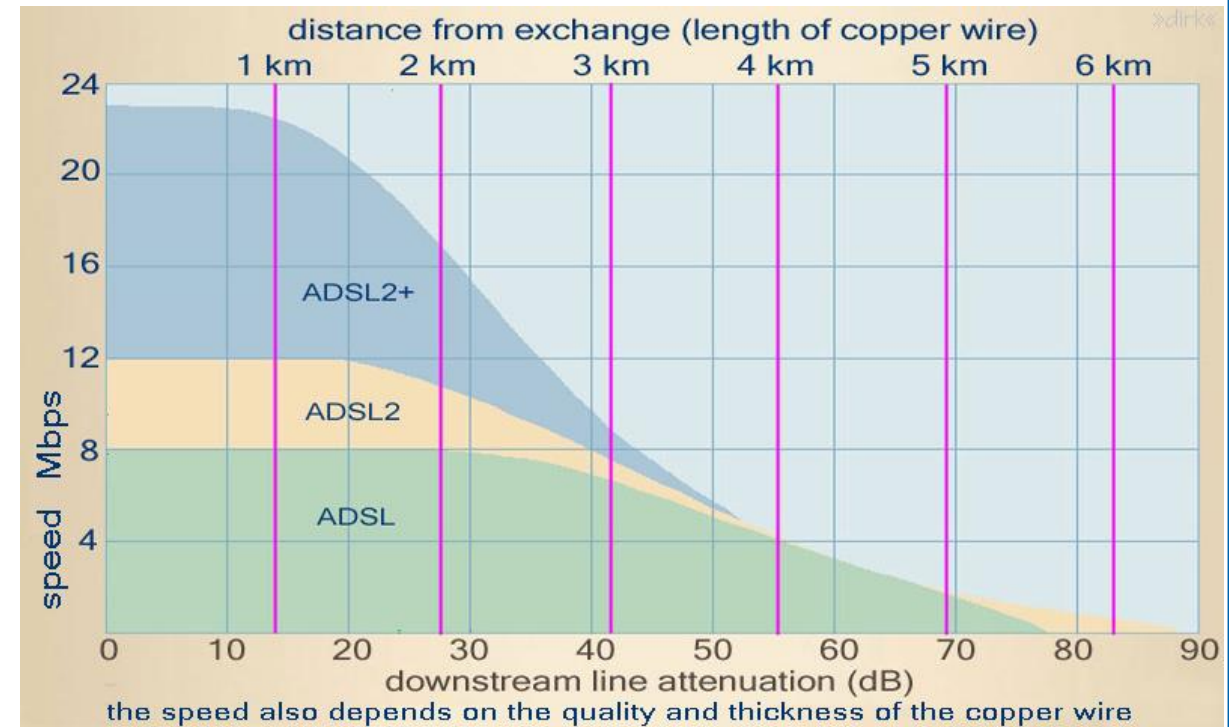
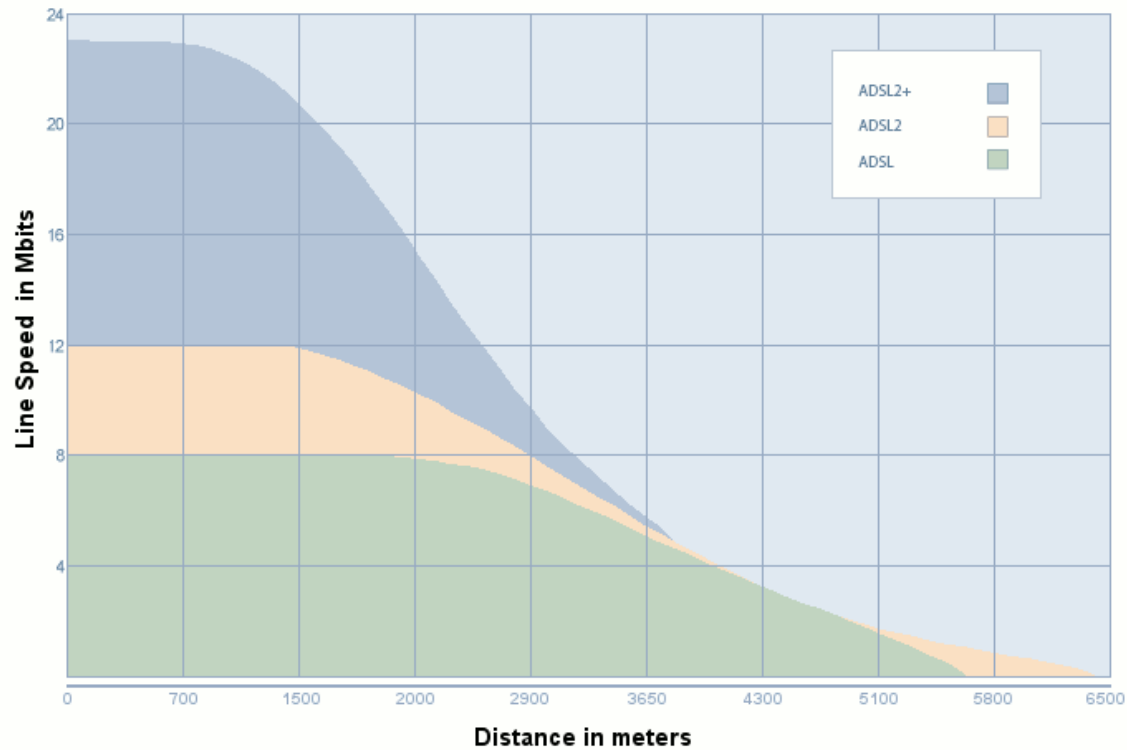


ADSL Technology

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- ADSL

How distance from the exchange affects maximum line speed



THANK YOU!

